

Fault Prognosis and Isolation of Lithium-Ion Batteries in Electric Vehicles Considering Real-Scenario Thermal Runaway Risks

Jichao Hong^{ID}, *Member, IEEE*, Zhenpo Wang^{ID}, *Member, IEEE*, Changhui Qu, Fei Ma, Xiaoming Xu, Jue Yang, Jinghan Zhang, Yangjie Zhou^{ID}, Tongxin Shan, and Yankai Hou

Abstract—Advanced safe battery storage systems with health prognostic performance are vital for electric vehicles. Various faults of lithium-ion batteries are usually undetectable in their early stage due to their concealment and graduality. This article presents a real-time fault diagnosis and isolation scheme for real-scenario batteries using the normalized discrete wavelet decomposition. The early frequency-domain features of the fault signals are extracted utilizing the high-frequency detail wavelet components, and a multilevel fault prognosis strategy is developed considering complex charging/driving characteristics under real-vehicle operating conditions. The verification results, implemented on loose wire connection batteries and real-scenario thermal runaway batteries, demonstrate that the proposed method can accurately extract and locate the hidden fault signals even under small magnitudes and effectively detecting and isolating battery faults before thermal runaway. Furthermore, significant reliability and stability of the proposed method are verified on more real-vehicle operation data, enabling online monitorable and traceable of battery faults before triggering thermal runaway, safeguarding drivers and passengers in real-world vehicular operation.

Index Terms—Battery storage system, discrete wavelet decomposition (DWD), electric vehicles (EVs), fault diagnosis, thermal runaway.

I. INTRODUCTION

A. Motivation and Incitement

ELECTRIC vehicles (EVs) will become a fundamental part of efficient and sustainable transportation systems

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Jichao Hong, Fei Ma, Xiaoming Xu, and Jue Yang are with the Department of Vehicle Engineering, School of Mechanical Engineering, University of Science and Technology Beijing, Beijing 100083, China, and also with the Shunde Graduate School, University of Science and Technology Beijing, Foshan 528000, China (e-mail: hongjichao@ustb.edu.cn; yeke@ustb.edu.cn; xuxiaoming3777@163.com).

Zhenpo Wang, Changhui Qu, Jinghan Zhang, Yangjie Zhou, Tongxin Shan, and Yankai Hou are with the National Engineering Laboratory for Electric Vehicles, School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China (e-mail: wangzhenpo@bit.edu.cn; quchanghui812@bit.edu.cn; zhangjinghanbit@outlook.com; zhouyangjie0801@gmail.com; shantongxin97@gmail.com; houyankai@163.com).

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TABLE I

COMPARISON OF THE EXISTING BATTERY FAULT DIAGNOSIS METHODS

Methods	Advantages	Disadvantages
Statistical thresholds	Fast real-time response	Inevitable alarm lag
Observer-based model	Stable model structure	Poor real-time response
Equivalent circuit model	High modeling accuracy	Poor practicability
Data-driven algorithms	Data adaptability	Limited calculation speed
Machine learning	Black-box learning	Huge data sample demand
Entropy algorithms	Hidden-signal recognition	Poor fault localization

in the future. As a key component, battery systems can largely determine the safety, drivability, and driving mileage of EVs [1]. A battery system composed of hundreds of cells in series and/or parallel configurations is, therefore, necessary due to the limited capacity and voltage of individual cells [2]–[4]. In spite of the continuing progress in battery manufacturing and packaging, battery faults may occur at each constituent or the associated accessories owing to the natural aging or abusive operation [5], [6]. In general, external abusive conditions, such as overcharging and short circuits, usually could lead to furious internal exothermic reactions inside battery packs [7]. When the heating rate exceeds the dissipation rate, the thermal runaway may occur, which gives rise to hazardous phenomena, such as smoking, gas venting, and even explosion [8]. However, due to the sensitivity and confidentiality of the accident data to the car companies, it is still difficult in obtaining accident data to perform verifications of the various diagnostic methods on the actual thermal runaway accident data, greatly hindering the advancement and development of high-safety battery systems in real-world vehicles. Table I shows the comparison of the existing battery fault diagnosis methods. Due to their inherent disadvantages and above-mentioned shortcomings, there is still much suffering in fault prognosis for real-scenario battery systems.

B. Literature Review

The common battery faults include faults of the battery itself, battery management system, sensors and connected components, and so on. The main body faults include gradual faults (capacity and power degradation) caused by battery aging and sudden faults (internal short circuit, thermal runaway, capacity dive, liquid leakage, and so on) that cause sudden failure of the power battery system or significant

performance degradation without obvious or short-term signs [9], [10]. Considerable efforts have been dedicated to traditional fault diagnosis and prognosis of battery systems. For instance, in order to improve computational efficiency and applicability in embedded systems, Yan *et al.* [11] introduced the Lebesgue Sampling for fault diagnosis of batteries by adopting the philosophy of “execution when necessary.” They designed and implemented a particle filter to directly pick certain model parameters, such as power fade, state of health, and so on. Hu *et al.* [12] employed the advanced sparse Bayesian predictive modeling to capture the underlying correspondence between the capacity loss and the sample entropy. The proposed approach allows for an analytical integration of temperature effects to establish an explicitly temperature-perspective health estimator, whose performance and complexity are contrasted to the support vector machine scheme. Hong *et al.* [13], [14] presented a novel fault diagnosis method for battery systems in EVs based on big data and machine-learning methods. Through machine learning training on real-vehicle operation big data, the multiforward-step prediction effect of multiple fault characteristic parameters of battery systems is obtained, and the early diagnosis of common battery abnormalities or faults based on the statistical threshold identification method can be realized. Xiong *et al.* [15] proposed a simple and effective model-based sensor fault detection method for lithium-ion batteries using the unscented Kalman filter. Via the synthesized design of Luenberger and learning Observers, Chen *et al.* [16] proposed simultaneous fault isolation and estimation scheme for Li-ion batteries. The fault-detection residuals can be generated and the fault sources can be located simultaneously by using this method.

Undetected faults would have detrimental effects on batteries and may lead to catastrophic accidents under some extreme conditions. One of the common faults that occur in battery cells is loose wire connections (LWCs). In battery packs, the cells’ wire connections are prone to becoming loose due to vibration and mechanical deformation during vehicular operations, thereby significantly increasing the contact resistance [17]. This may result in more severe accidents due to arcing and melting at cell contact points under high-voltage and high-current conditions. If undetected, they may lead to heating, burning, or even explosion. Thus, the fault detection and isolation of wire connections between cells are instrumental in preventing such latent hazards. However, there are a few related research studies in this area in recent years, and the existing literature mainly focuses on single-cell safety improvement by exploring appropriate electrolyte, electrode, and diaphragm materials [18]. In addition, the wire connection status of the battery is usually invisible, which makes it more difficult to discern the wire connection fault through conventional monitoring or detecting methods. Cuomo [19] designed a systematic method for detecting battery LWC failures. He calculated the difference between the cell’s temperature and connectors’ temperature at a measured current via a comparator. If the measured value exceeds the predetermined threshold, an alarm will be declared. However, this system has poor stability and is prone to false alarms. Yao *et al.* [20] proposed a method to detect the wire connection status of

Lithium-Ion batteries based on entropy. The experimental data shows that the ensemble Shannon entropy can predict the time and location of battery wire connection faults. However, this method is computationally intensive and has poor real-time monitoring performances. Xu *et al.* [21] used a first-order *RC* equivalent circuit model to analyze the connection failure of a battery pack. They determine the occurrence of contact-resistance fault by comparing voltage, current, and resistance before and after the fault. But, this method has too poor predicting performance to be predicted in time due to the hysteresis of the after-fault data.

In order to comprehensively carry out fault prognosis research of battery systems from a multidomain perspective, except the above-mentioned time-domain methods with inherent disadvantages, the frequency-domain-oriented features are also potential solutions to be developed. At present, the commonly used time-frequency conversion methods include Fourier transform [22], Laplace transform [23], and wavelet transform [24]. Fast Fourier transform (FFT) is not only an important tool for theoretical analysis and engineering applications but also has certain inherent limitations, such as the lack of localized analysis capabilities and the inability to analyze nonstationary signals. Wavelet analysis was proposed by Grossmann and Morlet [25]. It is a new field that is rapidly developing in applied mathematics and engineering based on Fourier analysis [26]. The wavelet transform inherits and develops the idea of localization of the short-time Fourier transform, and at the same time, overcomes the disadvantage that the window size does not change with frequency. Therefore, the wavelet transform is an ideal tool for time-frequency analysis and processing of nonstationary signals.

“Thermal runaway” refers to the overheating phenomenon caused by the exothermic chain reaction inside the battery, which leads to the rapid change of temperature rise rate of the battery. The battery accidents under extreme conditions, such as mechanical damage, electrical abuse, and thermal abuse, are all in the form of thermal runaway. Therefore, it is important to realize accurate fault diagnosis and early isolation for battery systems. In the early stage of thermal runaway, a large amount of gaseous material is produced by the side reaction inside the battery, and the change of temperature, voltage, and other characteristic parameters is still not obvious; however, limited by the sensor precision and packaging technology, the above-mentioned thermal runaway diagnosis technology cannot be applied in the battery system. The fault diagnosis and safety control of battery systems based on the real-vehicle operation big data has strong research and application value, but the research work carried out by scholars is relatively small at present. Hong *et al.* [14] and Jichao *et al.* [27] used the real-vehicle operation data, combined with entropy judgment theory, proposed a battery thermal runaway diagnosis method. Zhao *et al.* [28] proposed an outlier identification method based on direction propagation neural network machine learning algorithm and 3σ multilevel screening strategy for battery voltage faults and validated it in real vehicles, which provides guidance for the study of potential failure analysis of battery systems based on big data technology. However, the thermal runaway triggering causes

and influencing factors are diversified, the evolution law is complex, and the fault signals before the accident are highly concealed. Moreover, it has the characteristics of short reaction time, high danger, and strong coupling of characteristic parameters. Therefore, the existing fault diagnosis technology cannot realize the early identification and accurate isolation of the fault signals before the thermal runaway accident.

C. Contribution and Article Organization

The main purpose of this article is to provide a real-time prognosis and isolation scheme based on discrete wavelet transform (DWT) for battery faults in real-world EVs. Three notable contributions and improvements to the current techniques are attempted to be implemented as follows.

- 1) *Frequency-Domain-Based Signal Recognition*: This work proposes a fault prognosis scheme using the DWT-based discrete wavelet decomposition (DWD). The early hidden faults signals within the conventional time-domain-based threshold range can be extracted, and the original signal can be perfectly restored through time-frequency conversion, overcoming the defects of the single-diagnostic index, short alarm time using the traditional fault alarm method based on the critical threshold of key parameters and the change rate.
- 2) *Accurate Isolation for Fault Signals*: The proposed method can accurately identify and isolate fault signals through the high-frequency detail signals in real time, restoring the time and location of the battery fault, and enabling online monitorable and traceable of battery faults before thermal runaway.
- 3) *Real-Scenario Thermal Runaway Verification*: The real-vehicle operating data from a battery thermal runaway accident is derived for viability and reliability verifications. Taking into account the charging/driving characteristics, a multilevel prognosis strategy based on the normalized DWD (NDWD) method is developed, by which the real-time evaluation of the fault levels before battery thermal runaway can be performed.

The rest of this article is organized as follows. Section II describes the LWC fault and modes. Section III introduces the experimental platform and the prognosis method. Section IV validates the proposed NDWD-based fault prognosis effects for a real-world accident vehicle. Section V presents conclusions.

II. LOOSE WIRE CONNECTIONS AND MORE ANALYSIS

A. Description of Loose Wire Connection

Hundreds of cells are connected in a certain way through nuts or welding to form a battery pack. Due to the impact of accelerated stress on fittings, it is easy to cause the battery to beat during the long-time running of an EV, which will produce a lot of calories, be prone to sparks, and even lead to combustion, explosion, or other serious consequences. When a wire connection becomes loose in a battery system, the system voltage always exhibits two characteristics. First, it has no obvious voltage fluctuation in the initial period, which

makes it difficult to be discovered. Second, the deterioration of LWCs may be quickened and lead to spark in a few minutes.

The LWC in a battery system has great influences on running vehicles in the following three aspects.

- 1) *Safety*: The running vehicles are often in the state of acceleration, deceleration, or uniform running. This will constantly alternate the discharging and charging status of the battery system, making the current change dramatically. The battery system with LWC will lead to the fitting to rotate or beat with the vehicle's vibration. It will make the contact among battery cells change from surface contact to point contact, then the resistance will increase in a short time; the relative motion will happen between the connecting piece and cell at the same time, which is easy to produce sparks even fire. In addition, a battery system occurs LWC more often in the location of the contact joint between the connection point with the cell, but not the cells themselves. When the LWC occurs, the temperature of the cells themselves is not high and is hard to be detected by the battery management system.
- 2) *Dynamic Performance*: A battery system is the power source of the EV. If the internal connection is loose, the internal impedance will increase, and the inside local temperature of the battery system will occur mutations and produce a large number of calories. This will expand a part of battery energy and will shorten the travel mileage. At the same time, the local impedance of the battery system increases greatly, which also consumes the available energy of batteries and eventually leads to degrading the power performance.
- 3) *Degraded Performance of Estimation*: The battery management system uses batteries' terminal voltage and temperature at the measurement points. The voltage fluctuation is not obvious at beginning of LWC, and it will gradually become quite severe when vehicles are running; the voltage fluctuation will be accordingly bigger in terms of its magnitude, but the battery energy and state of charge are likely to be in a high-performance state [29]. This situation can eventually cause serious deviations when a battery's residual capacity is being estimated.

Even if the battery system has been carried on strict calibration and selection before loading, after a period of time, the connection among single cells will gradually become loose in battery systems. First, in the process of assembling all fittings of the battery system, there are certain processing stress and exist certain differences between each other of batteries. The stress cannot show itself in the stand-alone process, but the external load will impact fittings and make them lose with the practical operation of EVs. Second, during the operation of the vehicle, the battery system will be influenced by drivers' driving habits, bumps, and road conditions. The impacts on the fittings in different spaces have a certain discrepancy, which will exacerbate imbalanced distribution even cause the fittings to rupture, as shown in Fig. 1(a).

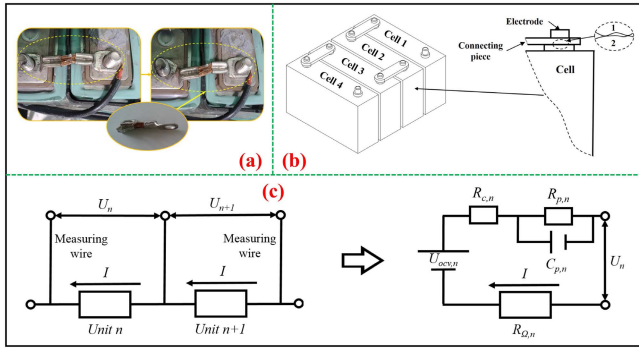


Fig. 1. Display of battery connection. (a) Rupture of fittings in a battery pack. (b) Connection method and a stationary electrical connection. (c) Connection mode and unit n .

In addition, certain differences in internal structures and the working environment in every single cell may cause different current densities on the fittings. Then, the working stress between fittings and single cells can become bigger and force fittings to be loose or breaking.

B. Analysis of Connection Modes

Fig. 1(b) shows a schematic presentation of the connection method and a stationary electrical contact, showing the connection method of four cells in a battery pack. The cells are connected in series, and the connection contains connecting pieces and measuring wires that are bolted to the cells' electrodes. A single cell and the connecting piece are combined to form a unit.

C. Modeling a Battery With Connection Considered

Based on the connection manner described in Fig. 1(b) and (c) further shows the connection mode and unit n , including a simplified and widely used Thevenin equivalent circuit model [30], where I is the current, $R_{\Omega,n}$ represents the internal resistance for unit n , $R_{c,n}$ is the contact resistance between a cell and the connecting piece, $U_{ocv,n}$ is the open-circuit voltage (OCV) for unit n as an ideal voltage source, $C_{p,n}$ is a polarization capacitance, U_n is the terminal voltage, and $R_{p,n}$ represents the sum of the concentration polarization impedance and electrochemical polarization impedance. Because the resistance of wires for measurements is much smaller than $R_{\Omega,n}$, as a result, the voltage drop through the measuring wire would be neglected and the measured voltage U_n could accurately predict the OCV; therefore, the relationship among these parameters can be formulated as follows:

$$U_n = U_{ocv,n}(t) - I(t) \times R_{\Omega,n} - U_{p,n}(t) - I(t) \times R_{c,n}. \quad (1)$$

The OCV is a continuous function of the state of charge for the battery system. The polarization voltage and the internal resistance are determined by the internal chemical substances, such as active materials at the positive and negative terminals of the battery; therefore, the change of the internal characteristics is not usually sudden but successful unless the battery is crashed or exploded. However,

Fig. 1(b) and (c) shows that the contact resistance is closely relevant to the connection status. If the pressure is too small to meet normal connection status, the contact area will have an obvious change and this change will affect the contact resistance.

As the preliminary analysis, the cell voltage measurement cannot cause the power fade, but the increased internal resistance or contact resistance can lead to the fault in terms of severe change of voltage in the battery pack. The fluctuation of terminal voltage is mainly due to the jump of the connection impedance. When the wire connection is in good condition, their magnitudes of the internal resistance and contact resistance are different and comparable [31]. The voltage change is mainly affected by the current, so it is difficult to distinguish between them at this time. Whereas, when the wire connection status begins to deteriorate, the magnitude of the contact resistance will change greatly. At this time, the terminal voltage of the battery will undergo a severe jump in a short time, accompanied by a local high-temperature phenomenon. Therefore, the voltage fluctuations depend mainly on the changes of the contact resistance in the event of LWCs, and the influence of the current on the voltage can be negligible at this time. In fact, the resistance is very difficult to be accurately monitored in real time and can only be properly estimated during the EVs running, so the voltage can be applied to diagnose connection status when the wire connection status deteriorates, which is much easier for real-time monitoring and calculation.

III. INTRODUCTION TO THE EXPERIMENTAL PLATFORM AND PROGNOSIS METHODS

A. Experimental Platform and Preliminary Analysis

In order to collect data for the fault in terms of LWC, an experimental test platform was set up, as shown in Fig. 2(a), which consists of a Digtron Battery Test System (BTS-600), a vibrating test bench, an information collector, several voltage sensors, and a host computer. The BTS-600 is assigned to charge/discharge the batteries according to a pre-designed loading profile with a maximum voltage of 21 V and a maximum constant charging/discharging current of 45 A, which is as shown in Fig. 2(b). The vibrating test bench is utilized to provide a vibration environment for the battery systems. The information collector is for logging the multichannel test data. Each of the tested Lithium-ion batteries has a nominal capacity of 50 Ah. Five battery cells are connected in series to form a small module. The voltage measurement is for four cells only while the fifth cell will not be taken as the research object.

Due to the vibration on the platform, there is severe noise corruption in the process of data collection. The discrete cosine transform [32] is used to filter the collected voltage data. The test of sweep-frequency vibration is conducted on a battery pack. When the charging/discharging starts in the condition of vibration, the voltage values will be collected by the information collector. After 30 charging/discharging cycles, the voltage curves in different loading cycles are obtained. The current and voltage curves of Cell 1 in the first cycle are

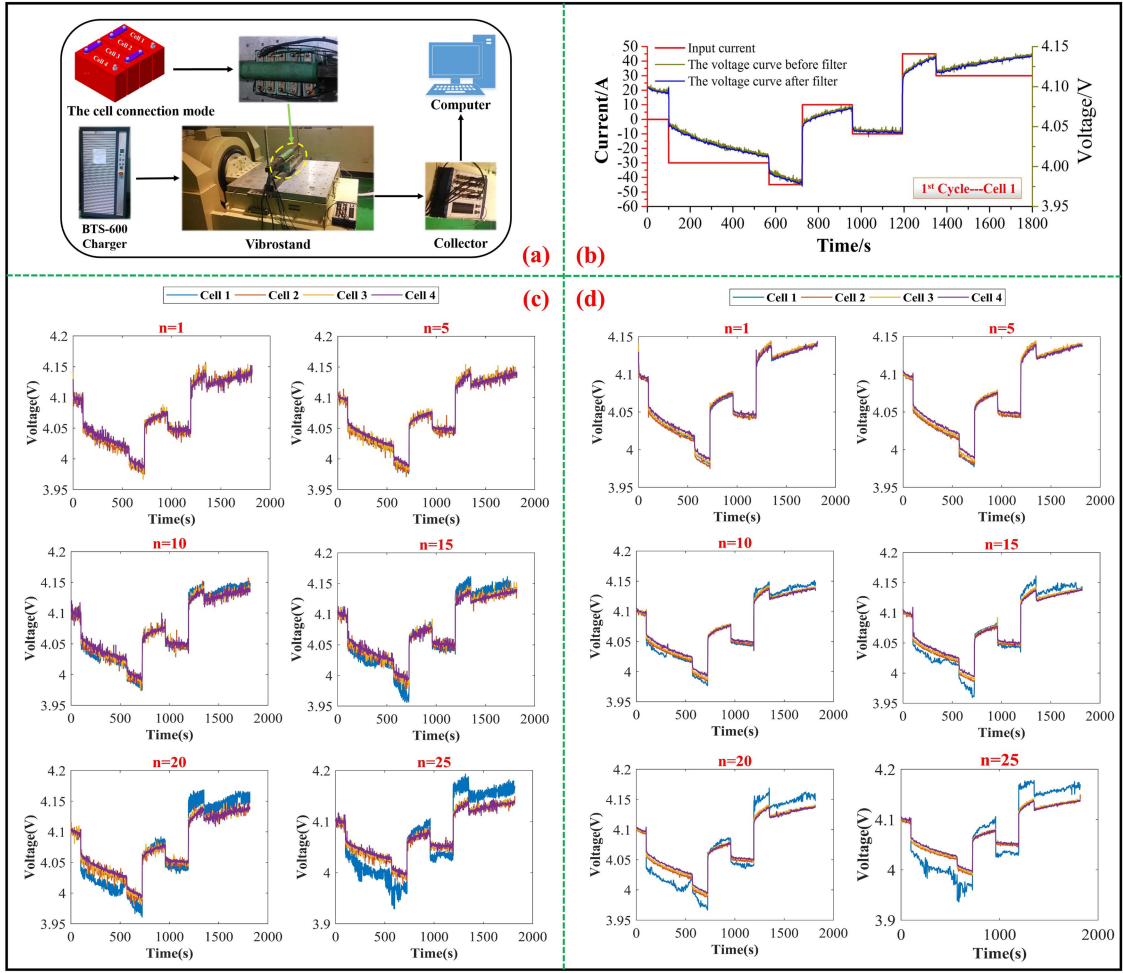


Fig. 2. Display of battery connection. (a) Experimental system of the battery wire connection fault. (b) Current and voltage curves of Cell 1 in the first cycle. (c) Voltage curves before filtering in different cycles. (d) Voltage curves after filtering in different cycles.

shown in Fig. 2(b). Fig. 2(c) and (d) shows the voltage curves before and after filtered in six cycles (1, 5, 10, 15, 20, and 25). It is clear that the deviation caused by system and measurement noises has been effectively filtered out, and the voltage difference becomes more apparent after filtering. The wire connections between cells are in good shape in the initial period, so the voltage fluctuation is not obvious. However, with time increasing, the voltage vibration curve becomes more prominent, which shows that the measured voltage curves have changed obviously. The deviation caused by system and/or measurement noises can be effectively filtered, and the true voltage values can be retained.

The measured voltage values, as shown in Fig. 2(d), indicate that there are no obvious fluctuations in the condition of constant-current loadings, such as Cycle 1 and Cycle 5. However, in Cycle 10, cell 1's voltage endures obvious fluctuations, and this becomes more evident with the continuing charging/discharging. Meanwhile, a nut is loosened. In Cycle 20, the voltage fluctuation of cell 1 is far higher than that of the other three cells with sparks occurring, indicating that the wire connection between the batteries is losing and this wire connection fault is more and more severe.

B. Introduction to the Prognosis Methods

1) *FFT*: FFT is a discipline that was first proposed by Cully and Tukey [33]. It is the fast algorithm for discrete Fourier transform (DFT) operation, including the Fourier series and the Fourier Transformation. FFT is a reversible transform, that is, it allows conversion between the original signal and the transformed signal. However, this kind of transformation has only one kind of information available at any time. Because the frequency domain after FFT does not contain the original time information, the time information before the transformation cannot be obtained in the time domain after the inverse transformation. In fact, this phenomenon can be understood in combination with the formula of the Fourier transform. The Fourier transform can be divided into continuous Fourier transform and DFT.

The formula for the continuous Fourier transform is

$$F(w) = F[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-iwt} dt \quad (2)$$

$$f(t) = F^{-1}[F(w)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(w)e^{iwt} dw. \quad (3)$$

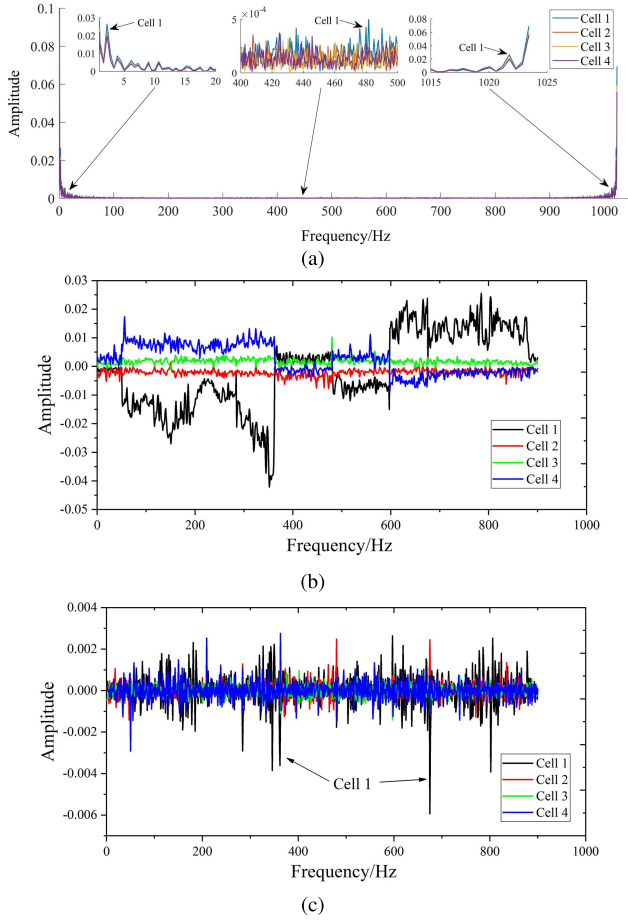


Fig. 3. Cycle 20 after FFT and DWD. (a) Amplitude-frequency signals after FFT. (b) Low-frequency signals after DWD. (c) High-frequency signals after DWD.

The formula for the DFT is

$$X[k] = \sum_{n=0}^{N-1} x_n e^{-i2\pi kn/N} \quad (4)$$

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{i2\pi kn/N} \quad (5)$$

where $F(w)$ is the mirror function of $f(t)$, $f(t)$ is the original mirror function of $F(w)$, $X[k]$ represents the transformed data, x_n is the sampled analog signal, and N is the number of sampling points.

We can see that because the integral in the Fourier transform is from negative infinity to positive infinity, this integral is valid for all time classes, that is, the globality of the integral is affected no matter when the frequency component changes, and the same is true of the DFT, so this is why the FFT is usually not suitable for analyzing nonstationary signals. Various parameters of a battery system, especially the fault signal, must be some nonstationary signals, so the FFT has no advantage in the frequency domain analysis of the battery system. For instance, we take Cycle 20 in the virtual connection failure as an example, and the discrete FFT result is shown in Fig. 3(a). From Fig. 3(a), we can see that whether in the low-frequency,

intermediate-frequency, or high-frequency phases (which are, respectively, 0~20, 400~500, and 1015~1024 HZ), the result of the time-frequency conversion actually makes the abnormality of the virtual Cell 1 less obvious, that is, the FFT-based anomaly diagnosis is not effective.

2) *DWD*: Wavelet, as its name implies, is a small waveform that has both volatility and attenuation characteristics. Compared with the Fourier transform, the wavelet transform has the following advantages: first, it can analyze the local characteristics of the signals; second, it can simultaneously analyze the signals in the time and frequency domain; and third, it can analyze the nonstationary signal with effective noise reduction [34]. Furthermore, the wavelet transform can focus on any detail of the signal by using gradually fine sampling steps for different frequency components, so it has good time-frequency localization characteristics and handles weak or abrupt signals. What needs to be emphasized is that it is especially suitable for signal analysis of the nonsteady-state distortion waveform problem.

Arbitrary signal or function $f(t)$ is defined in the function space or signal space $L^2(R)$, and the expansion in the continuous wavelet base $\varphi_{(\alpha,\tau)}(t)$. $WT_f(\alpha, \tau)$ is called continuous wavelet transform of $f(t)$

$$WT_f(\alpha, \tau) = \frac{1}{\sqrt{|\alpha|}} \int_R f(t) \varphi_{(\alpha,\tau)}(t) dt \quad (6)$$

where

$$\varphi_{(\alpha,\tau)}(t) = \frac{1}{\sqrt{|\alpha|}} \varphi\left(\frac{t-\tau}{\alpha}\right) \quad (7)$$

$$C_\varphi = \int_R \frac{\varphi(v)}{|v|} dv < +\infty. \quad (8)$$

The scale parameter α and translation parameter τ determine the time-frequency characteristics of the wavelet transform. Because the coefficients of the continuous wavelet transform are coupled and repetitive, they need to be further discretized in the actual application process, that is, the DWT and the discretized parameters, such as α and τ , are obtained. The steps in discrete mode are as follows.

1) Power-level discretization of scale parameter α

$$\alpha = \alpha_0^m. \quad (9)$$

2) Discretization of translation parameter τ

$$\tau = k\alpha_0^m \tau_0 \quad (10)$$

where $\alpha > 0$, $\tau > 0$, m , and k are all integers.

After the scale parameters and translation parameters are discretized, the discrete wavelet basis $\varphi_{\alpha_0^m, k\tau_0}$ and DWT $WT_f(\alpha_0^m, k\tau_0)$ can be obtained as follows:

$$\varphi_{\alpha_0^m, k\tau_0} = \alpha_0^{-\frac{m}{2}} \varphi(\alpha_0^{-m} t - k\tau_0) \quad (11)$$

$$WT_f(\alpha_0^m, k\tau_0) = \alpha_0^{-\frac{m}{2}} \int_R f(t) \varphi(\alpha_0^{-m} t - k\tau_0) dt. \quad (12)$$

The basis of the FFT is a sine-cosine function, while the basis of a wavelet transform is a wavelet of various shapes, that is, it treats the entire waveform as a superposition of wavelets with different positions and widths. The scale

parameter α controls small expansion and contraction, and the translation parameter τ can control the wavelet translation. According to the above analysis, the FFT is not suitable for processing nonstationary signals and can only obtain a frequency spectrum, and the frequency-domain information only provides the waveform components but does not provide the component appearance order; therefore, the simple Fourier analysis has a poor resolution for nonstationary signals whose frequency varies with time.

Wavelet analysis can obtain a time-frequency spectrum, including signal decomposition (i.e., wavelet decomposition) and reconstruction (i.e., wavelet transform). This makes wavelet analysis very suitable for analyzing nonstationary time series, such as real-world battery signals, in this study. Wavelet decomposition is to find the best mapping from the actual signal space to the wavelet function space in order to obtain the best recovery of the original signal. The low-frequency and high-frequency signals obtained after the decomposition can perfectly restore the original signal by reconstruction, so it is widely used for denoising of various signals, such as images [35], [36]. In addition, the high-frequency and low-frequency components are obtained after wavelet decomposition. The low-frequency components obtained from the low-pass filter can be called “approximate components,” and the high-frequency components obtained from the high-pass filter can be called “detail components.” The position of the component corresponds to the original signal, so the signal after wavelet decomposition can perfectly reconstruct and locate the details in the original signal, and the coefficients of each layer of the wavelet decomposition can be quantified by selecting an appropriate threshold. A mother wavelet is determined to build the high- and low-pass filters, and the input signal is convolved with the two filters formed, respectively, and then downsampled to obtain the approximation and detail coefficients, which correspond to the low- and high-frequency coefficients.

The DWD can have many layers. The first layer is the decomposition of the signal into two parts, low-frequency information and high-frequency information, by wavelet analysis. The low-frequency information is the slow-changing part, which is the outline of the change in the data and accounts for most of the total information; the high-frequency information is the fast-changing part, which reflects the detailed information of the signals and accounts for a small part of the total information. On the basis of the first layer, the high-frequency information can be decomposed further into low-frequency and high-frequency parts, and the third layer is to decompose the high-frequency information from the second layer into low-frequency and high-frequency parts, and so on. The above is a single-scale decomposition, and the low-frequency part is not decomposed again. In the case of multiscale decomposition, the low-frequency part is decomposed one layer at a time, just like the high-frequency part. In this study, we use only one layer of decomposition.

Cycle 20 in the virtual connection failure is still taken as the processing object, the DWD results of in low frequency and high frequency are shown in Fig. 3. From Fig. 3(b), we can see that the low-frequency signal after wavelet decomposition

basically retains the characteristics of the original signals. The high-frequency signal, as shown in Fig. 3(c), can effectively display the location and amplitude of the abnormal signals and sudden changes of Cell 1. Therefore, we will further explore the processing effect of DWD on actual vehicle data under more complicated dynamic conditions.

IV. NDWD-BASED FAULT PROGNOSIS FOR A REAL-WORLD VEHICLE

A. Introduction to the Thermal Runaway Accident

Vehicle accidents caused by battery thermal runaways have been reported around the world in recent years, and the safety of EVs has received great attention from governments, vehicle manufacturers, and researchers. The safety concerns of EVs, if not resolved in a timely manner, will reduce consumer confidence in buying and trigger depression in the new energy vehicle market. The studied vehicle data come from the National Monitoring and Management Platform for EVs (NMMP-EV), which has been able to monitor and manage more than three million new energy vehicles across China in real time. Besides, the platform not only carries out statistical analysis work on monitored vehicles but also performs fault management for battery systems. The monitoring and management mechanism of the NMMP-EV Platform is demonstrated in Fig. 4(a). At present, the vehicle data containing battery thermal runaway accidents has been connected to the platform, in the hope that through accident investigation and further research to explore the cause of the accident and develop the real-time prognosis function for battery fault or thermal runaway for real-world EVs.

The studied vehicle is a pure EV with a spontaneous combustion accident that occurred in the second half of 2019 (based on corporate privacy, the following accident time and some other information have been desensitized). Spontaneous combustion occurs after driving a mileage after being fully charged on the accident day. The car is equipped with a ternary lithium battery pack with 91 single cells assembled in series using soft pack laminations. Fig. 4(b) shows the schematic of the distribution and different damage locations of the studied battery pack. A and B represent the rear and front of the battery pack, C and D are the right and left sides of the battery pack. It can be seen from Fig. 4(b) that the right and rear parts of the battery pack are basically intact after the accident, but the front part is perforated, and the other parts on the left are basically intact but seriously damaged at about 1/3 position.

The battery voltage curves of 15 days before the thermal runaway are shown in Fig. 5. As shown in Fig. 5, the car was charged only once on the accident day, and a thermal runaway occurred just a few minutes after it started driving. The voltages of Cell 9, Cell 7, and Cell 13 first dropped rapidly, and then Cell 23, Cell 19, and Cell 5 also gradually dropped. This also reflects the starting position and diffusion direction of the thermal runaway, which is consistent with the cell distribution in Fig. 4(b) (Cell 7 and Cell 13 are both concentrated near Cell 9). In addition, we found that during the 15 days before the thermal runaway occurred, the car underwent

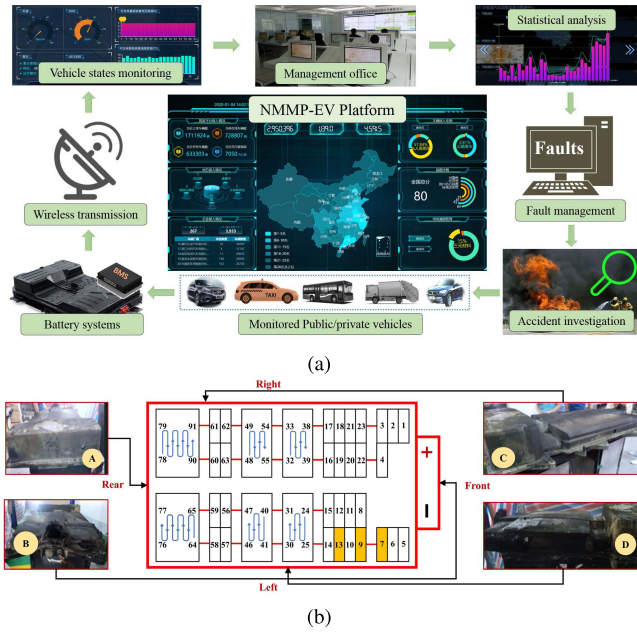


Fig. 4. Platform and incident descriptions. (a) Monitoring and management mechanism of the NMMP-EV platform. (b) Distribution and different damage locations of the studied battery pack.

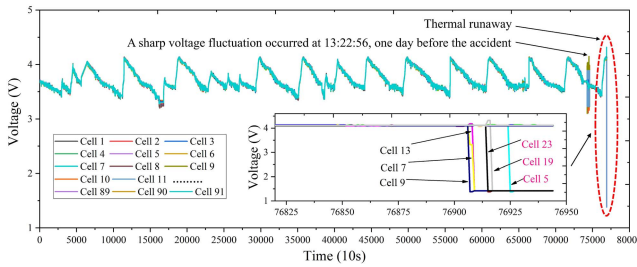


Fig. 5. Battery voltage curves before the accident.

about 20 driving/charging cycles with different depths, and the voltages fluctuations during the charging phase and the driving phase were significantly different. In addition, after 13:22:56 AM on the day before the accident, a short-time sharp voltage fluctuation occurred while the vehicle was driving. It is important to emphasize that this mutation may be caused by various reasons, such as the sudden deterioration of the health status of the studied battery system before it is close to thermal runaway, some unexpected operation situations, data acquisition error, or too rough data preprocessing. However, even if there are some errors in the collected data, there only is an influence on the authenticity of the original data of the studied vehicle, but no interference to the effectiveness of the proposed method and the generality of the application object.

Because the voltage characteristics between the charging state and the driving state during the actual vehicle operation are extremely different, we divide the actual vehicle operation data into driving data and charging data, and select some specific date of charging/driving data for analysis. Fig. 6(a) shows the 1-h charging voltage curves on the accident day,

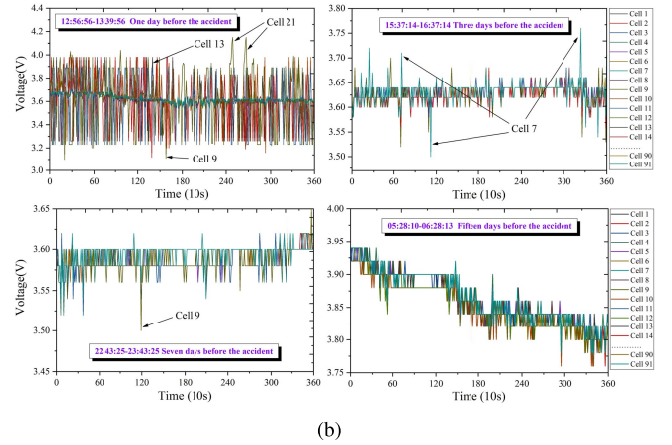
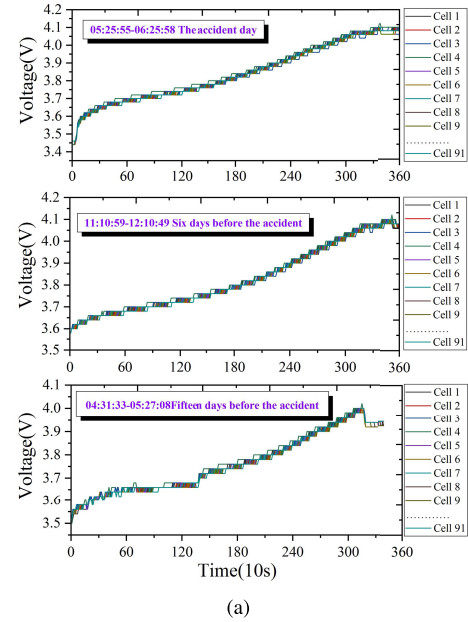


Fig. 6. Charging/driving voltage curves before the accident. (a) Charging voltage curves of three days before the accident. (b) Driving voltage curves of four days before the accident.

six days before the accident, and 15 days before the accident. Fig. 6(b) shows the 1-h driving voltage curves one day, three days, seven days, and 15 days before the accident, respectively (there is only a small amount of charging data without driving data on the accident day and six days before the accident).

Fig. 6(a) shows that the fluctuations of all the cell voltages in the three charging conditions are very smooth and the consistency is very good. Fig. 6(b) demonstrates that the cell voltages in the four groups of driving conditions have very sharp fluctuations due to the complex driving conditions. Moreover, one day, three days, and seven days before the accident, especially one day before the thermal runaway accident, Cell 7, Cell 9, and Cell 13, have more obvious abnormal fluctuations than other cells. This is consistent with the accident investigation results in Figs. 4(b) and 5. However, the fluctuations of all cell voltages are still within the time-domain-based safety threshold; therefore, it is difficult to accurately diagnose these abnormal cells with conventional methods.

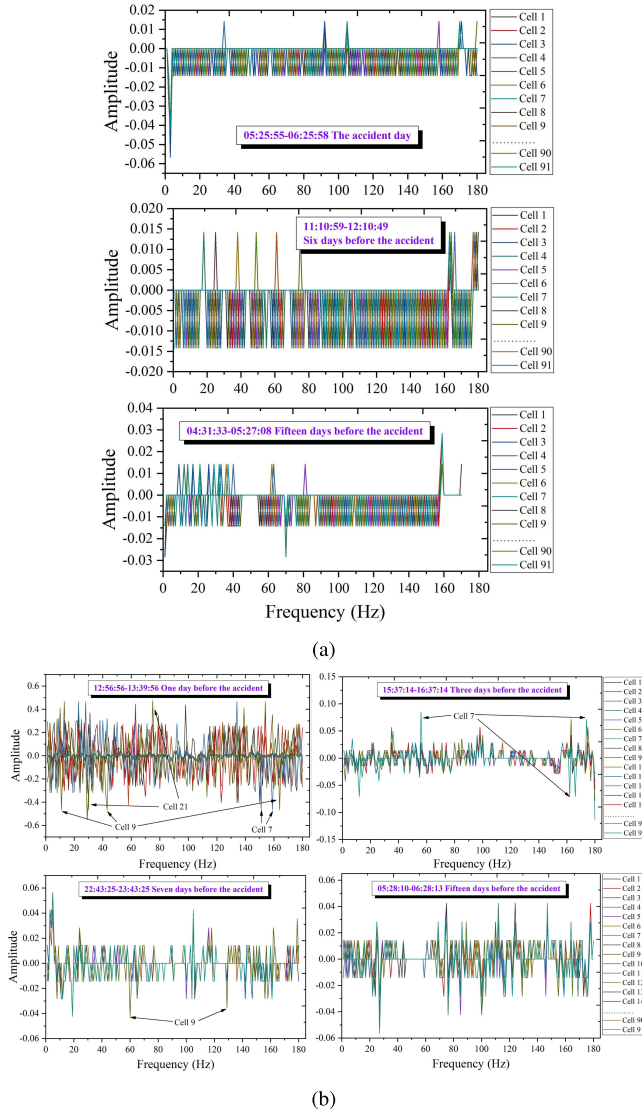


Fig. 7. Charging/driving DWD curves before the accident. (a) Charging DWD curves of three days before the accident. (b) Driving DWD curves of four days before the accident.

B. NDWD-Based Fault Prognosis Strategy

Fig. 7(a) and (b) are the high-frequency components of the DWDs of the above-mentioned three charging phases and four driving phases. Fig. 7(a) indicates that the DWD curves of all cell voltages in different charging phases have small fluctuations and good consistency, which corresponds to the voltage curves in Fig. 6(a). Due to the dynamic variability of driving conditions, the DWD curves in the four driving phases before the accident in Fig. 7(b) have sharp fluctuations. There are abnormal fluctuations in Cell 7 and Cell 9 three days and seven days before the accident, respectively; Cell 7, Cell 9, and Cell 13 have obvious abnormal fluctuations one day before the accident, which are also matched with the voltage curves in Fig. 6(b). Therefore, the proposed DWD method is very promising to effectively and stably diagnose the cells with abnormal voltages.

Based on the above analysis, we can find that the characteristics of battery voltages and DWDs are very different

due to different operating conditions between the charging and driving phases. Taking advantage of the big-data-based empirical methods of actual vehicles from the NMMP-EV platform, aiming at the operating characteristics of EVs, we developed a multilevel NDWD-based prognosis strategy combined with the normalization method. Through verification of a large number of real-world vehicle data and the trial-error method, anomaly coefficient thresholds for driving and charging states are correspondingly obtained, as shown in Fig. 8.

However, the DWD values of different battery types under different driving conditions are not fixed within a certain range. In order to obtain the thresholds of the abnormal coefficients with generality, the DWDs under the driving state are normalized from the beginning, then the first-level alarm threshold $T_{d1} = 0.8$ and the second-level warning threshold $T_{d2} = 0.6$ are determined. Besides, the vehicle always works in a stable and good-consistency state under the charging state; therefore, a generic abnormal coefficient threshold is determined as $T_c = \pm 0.015$. Through extensive verification of actual vehicle data, the determined thresholds perform great versatility and diagnostic accuracy. The rules for making anomaly coefficients are as follows.

- 1) The anomaly coefficient under driving state is

$$DWD_{d_nor} = \frac{|DWD_d| - |DWD_{d_min}|}{|DWD_{d_max}| - |DWD_{d_min}|} \quad (13)$$

$$A_D = DWD_{d_nor} - DWD_{d_ave} \quad (14)$$

where DWD_{d_min} is the minimum DWD of cell voltage, DWD_{d_max} is the maximum DWD of cell voltage, DWD_{d_nor} is DWD after normalization, DWD_{d_ave} is the average DWD of cell voltage, and A_D is the driving anomaly coefficient.

- 2) The anomaly coefficient under charging state is

$$A_C = DWD_c - DWD_{c_ave} \quad (15)$$

where DWD_{c_ave} is the average DWD of cell voltage, and A_C is the charging anomaly coefficient.

V. RESULTS AND DISCUSSION

Based on the abnormal coefficients set earlier, we performed verification on the studied vehicle. The abnormal coefficient curves of three charging states and four driving states are obtained, as shown in Fig. 9(a) and (b), respectively. Fig. 9(a) shows that the three sets of charging anomaly coefficients are always between -0.015 and 0.015 , that is, the battery system is in a safe charging state; this is consistent with the charging voltage characteristics in Fig. 9(a). Fig. 9(b) indicates that all cells are safe 15 days before the accident; the abnormality coefficients of Cell 7 and Cell 9 exceed the second-level warning threshold three days and seven days before the accident, respectively; Cell 7, Cell 9, and Cell 13 reach or exceed the first-level alarm threshold one day before the accident, when is very close to the occurrence of battery thermal runaway the next day. Therefore, based on the NDWD-based multilevel prognosis strategy, the abnormal Cell 9 can be detected and isolated at least seven days before the thermal runaway accident according to a second-level warning. More importantly,

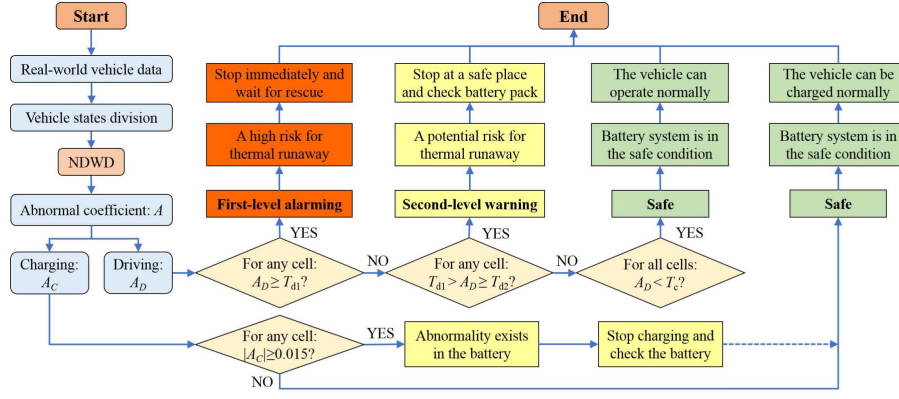


Fig. 8. NDWD-based multilevel fault prognosis strategy for real-world vehicles.

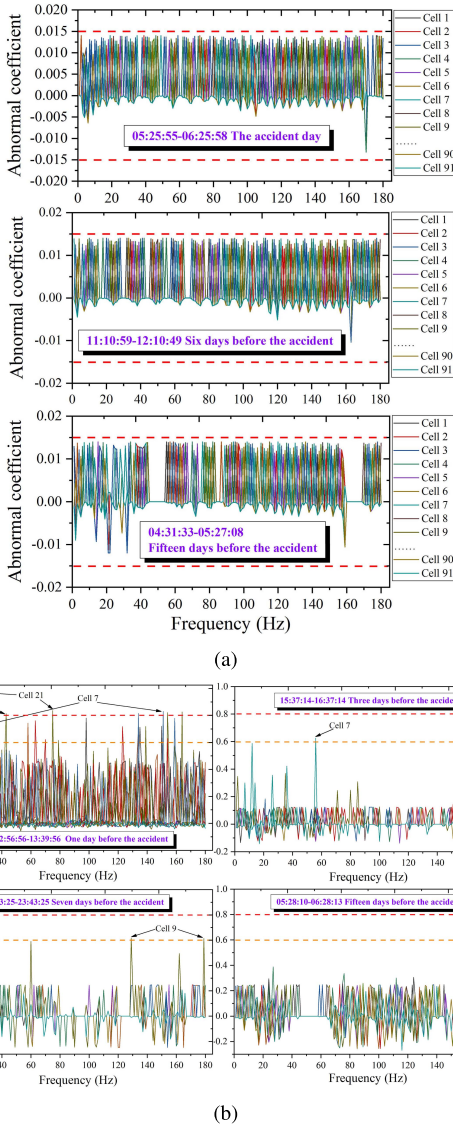


Fig. 9. Charging/driving anomaly coefficients before the accident. (a) Charging anomaly coefficients of three days before the accident. (b) Driving anomaly coefficients of four days before the accident.

through reasonable and early inspection, repair, or replacement of abnormal cells, the thermal runaway accident can be timely avoided to safeguard the drivers and passengers.

At present, the accident data of new energy vehicle accidents are still in the confidential stage, and the quality of the actual vehicle operation data is often very poor, so it is very difficult to obtain effective research data, and it is even rarer and valuable to find accidents caused by an external short circuit. In the near future, we will mainly carry out two aspects of work. On the one hand, based on the NMMP-EV platform, we will collect more data from accident vehicles caused by an external short circuit to further verify the feasibility and stability of the proposed method. On the other hand, we will collect data from accidents caused by other reasons, such as internal short circuits, overcharging, and so on, to verify the robustness and universality of the proposed method.

VI. CONCLUSION

This article presented a real-time prognosis and isolation scheme for battery faults based on the NDWD algorithm, which can effectively extract battery fault signals at an early stage. The high-frequency detail components of wavelet decomposition are used to extract the frequency-domain characteristics of the early fault signals. The verification results of experimental-based battery connection faults showed that the proposed method can effectively extract and locate the early hidden fault signals within the conventional time-domain-based threshold range. In addition, considering the different characteristics of the driving state and charging state of the battery system during actual vehicle operation, a multilevel fault prognosis strategy is developed based on the NDWD. The verification results of a real vehicle with a thermal runaway accident show that the proposed prognosis strategy can effectively detect and locate fault cells seven days before the thermal runaway is triggered. Through reasonable inspection, repair, or replacement measures, battery thermal runaway can be avoided early, thereby safeguarding the drivers and passengers.

Moreover, the reliability and stability of the method have been verified through the verification of multiple sets of real-world vehicle data. More importantly, this is an innovative battery fault prognosis research performed on an actual accident vehicle, proving its practical potential in the safety management of real-world battery systems. In the near future, more actual vehicle data of more types of battery faults

(before and after thermal runaway accidents) will be accumulated to verify and perfect the prognosis strategy. Combined with the time-domain-based diagnostic technologies, multidomain and multiscale battery fault prognosis can be realized to enables online monitorable and traceable battery faults before thermal runaway in real-world EVs.

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Jichao Hong (Member, IEEE) received the M.S. degree in automotive engineering from Qingdao University, Qingdao, China, in 2015, and the Ph.D. degree in automotive engineering from the Beijing Institute of Technology, Beijing, China, in 2020.

He is currently an Associate Professor with the University of Science and Technology Beijing, Beijing. His research interests include big data mining and analysis of new energy vehicles and power battery system integration and safety control.



Zhenpo Wang (Member, IEEE) received the Ph.D. degree in automotive engineering from the Beijing Institute of Technology, Beijing, China, in 2005.

He is currently a Professor with the Beijing Institute of Technology, where he is also an Associate Director with the National Engineering Laboratory for Electric Vehicles. His current research interests include pure electric vehicle integration, packaging and energy management of battery systems, and charging station designs.



Changhui Qu received the M.S. degree from Northeastern University, Shenyang, China, in 2013, and the Ph.D. degree in automotive engineering from the Beijing Institute of Technology, Beijing, China, in 2020.

He is currently a Post-Doctoral Researcher with the Beijing Institute of Technology. His current research interests include pure electric vehicle integration and safety management of battery systems.



Jinghan Zhang received the M.S. degree in automotive engineering from the Hebei University of Technology, Tianjin, China, in 2019. He is currently pursuing the Ph.D. degree with the Beijing Institute of Technology, Beijing, China.

His research interests include big data mining and analysis of new energy vehicles and power battery system research and development.



Fei Ma received the Ph.D. degree in automotive engineering from the University of Science and Technology Beijing, Beijing, China, in 2005.

He is currently a Professor with the University of Science and Technology Beijing, where he is also the Dean of the School of Mechanical Engineering. His current research interests include vehicle system dynamics, vibration and noise analysis and control, and self-vibration cavitating jet technologies.



Yangjie Zhou received the B.S. degree in mechanical engineering from the Beijing Institute of Technology, Beijing, China, in 2019, where he is currently pursuing the Ph.D. degree in mechanical engineering with the National Engineering Laboratory for Electric Vehicles.

His research interests include electric vehicles energy storage system safety, hazard prognosis strategies, and hazard control technologies.



Xiaoming Xu received the Ph.D. degree in automotive engineering from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2012.

He is currently a Professor with the University of Science and Technology Beijing, Beijing, China. His current research interests include new energy vehicle power battery systems, including Li-ion battery and fuel cell, and energy storage technologies.



Tongxin Shan received the B.S. degree in ammunition engineering and explosive technology from the Beijing Institute of Technology, Beijing, China, in 2019, where she is currently pursuing the Ph.D. degree in mechanical engineering with the National Engineering Laboratory for Electric Vehicles.

Her research interests include energy storage system safety and its risk prevention and hazard control technologies of electric vehicles.



Jue Yang received the Ph.D. degree in automotive engineering from the University of Science and Technology Beijing, Beijing, China, in 2005.

He is currently a Professor with the University of Science and Technology Beijing, where he is also the Head of the Department of Vehicle Engineering. His current research interests include vehicle intelligent control technology and vehicle system dynamics.



Yankai Hou was born in China, in 1996. He received the M.S. degree in automotive engineering from the Beijing Institute of Technology, Beijing, China, in 2021.

His research interests include modeling and state of health estimation of power batteries based on real-world big data.